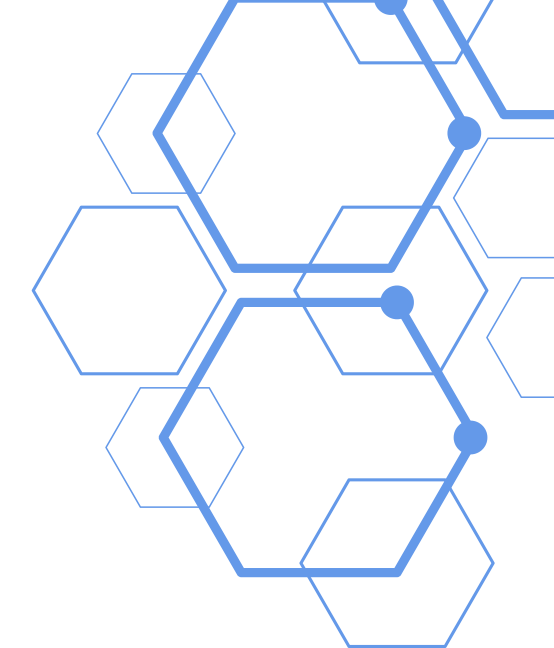
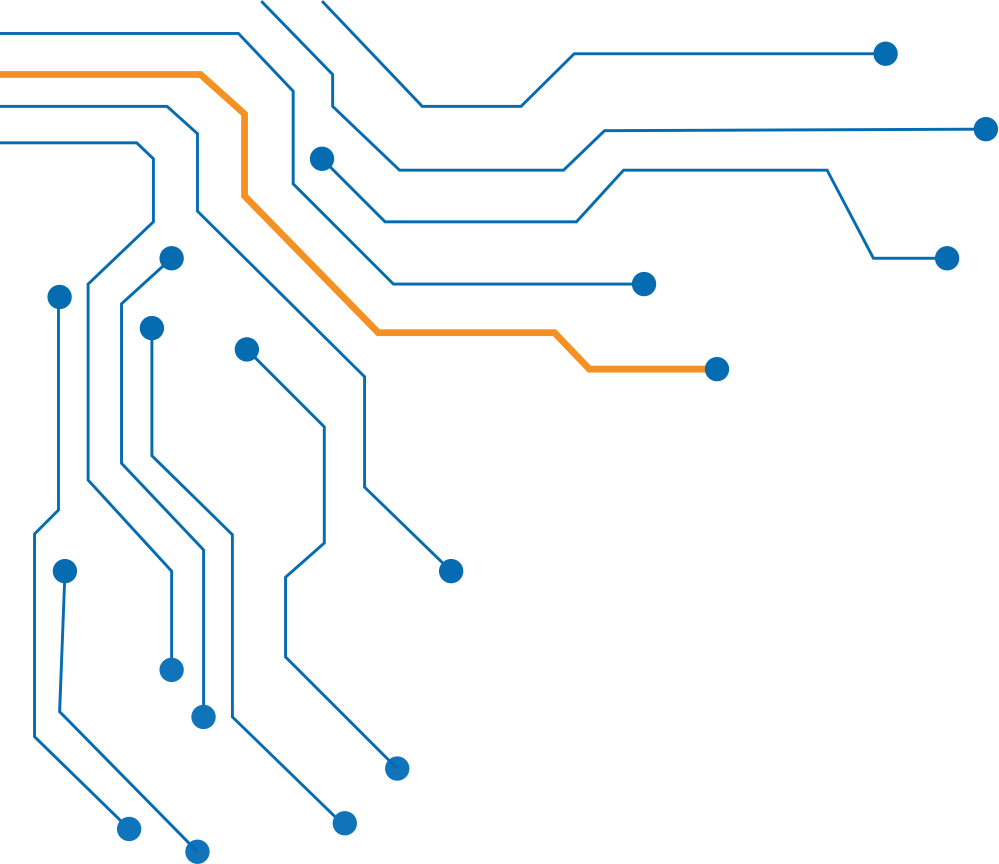




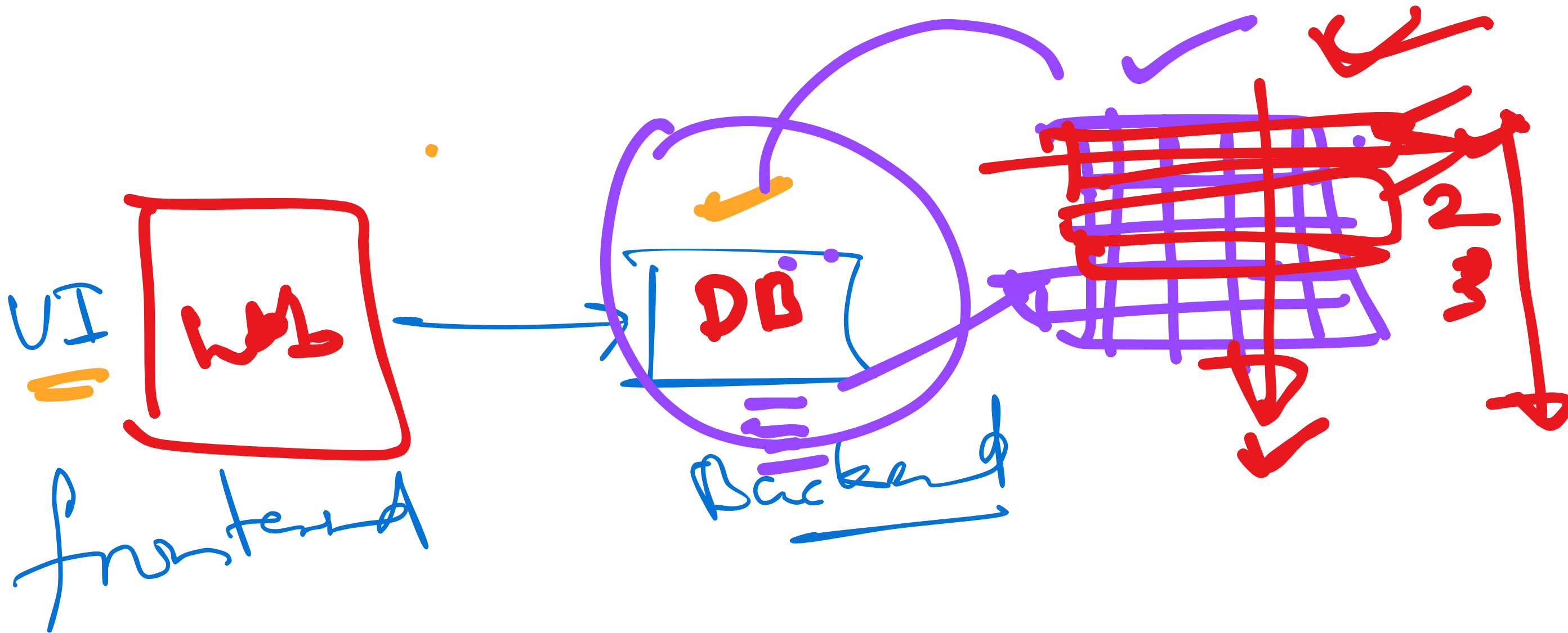
DATAWARE HOUSE CONCEPTS

GENERAL TERMINOLOGIES IN DATABASE AND DATA WAREHOUSE



TABLE

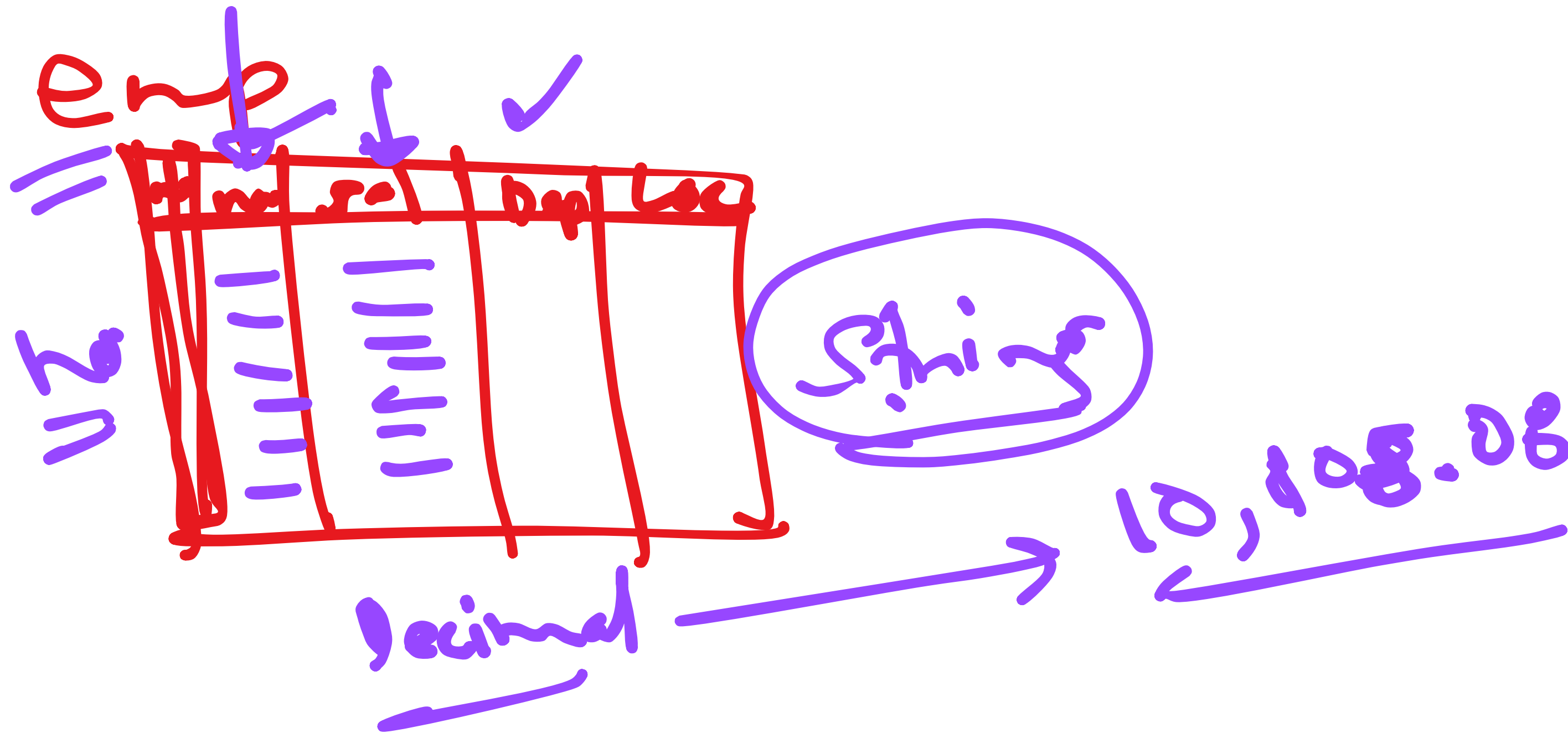
A table is a database object that stores data in rows and columns.
It can be understood as a collection of records.



COLUMN (ATTRIBUTE)

A column defines one specific type of data in a table.

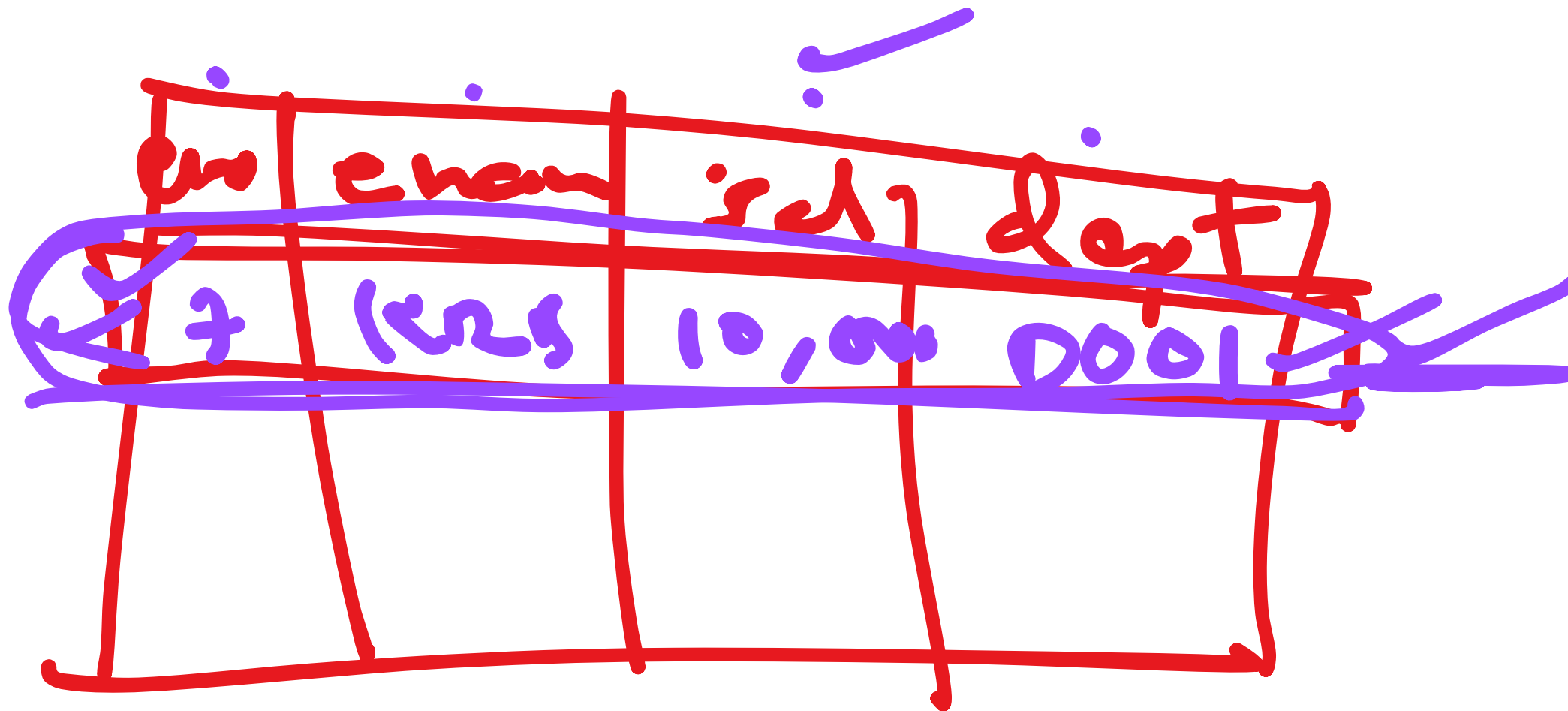
It contains a set of values of the same data type.



ROW (RECORD)

A row is a collection of fields that represents one complete set of information.

Each row contains values for different columns.



7	K23	10,00	DOOL



FIELD ✓

A field is a single piece of information stored inside a column.

Example: If “Flower” is a column, then Rose, Lily, Lotus are the field values.

Flower

Name	Type	Reg.
Rox	Red	Blor

METADATA

Metadata means “data about data.”

It gives higher-level information about the structure and meaning of the data.

Metadata

String(100)	Decimal(6)	String(1)
Name	Sal	Dept
Krishna	10,000	D001
Ram	11,100	D002
Jupreet	15,100	D003

METADATA SHEET

A metadata sheet is a document (Excel or tool-generated) that contains important details about a table and its columns.

It usually contains:

- **Table Name** ✓
- **Column Name** ✓
- **Data Type** ✓
- **Comments / Description** ✓

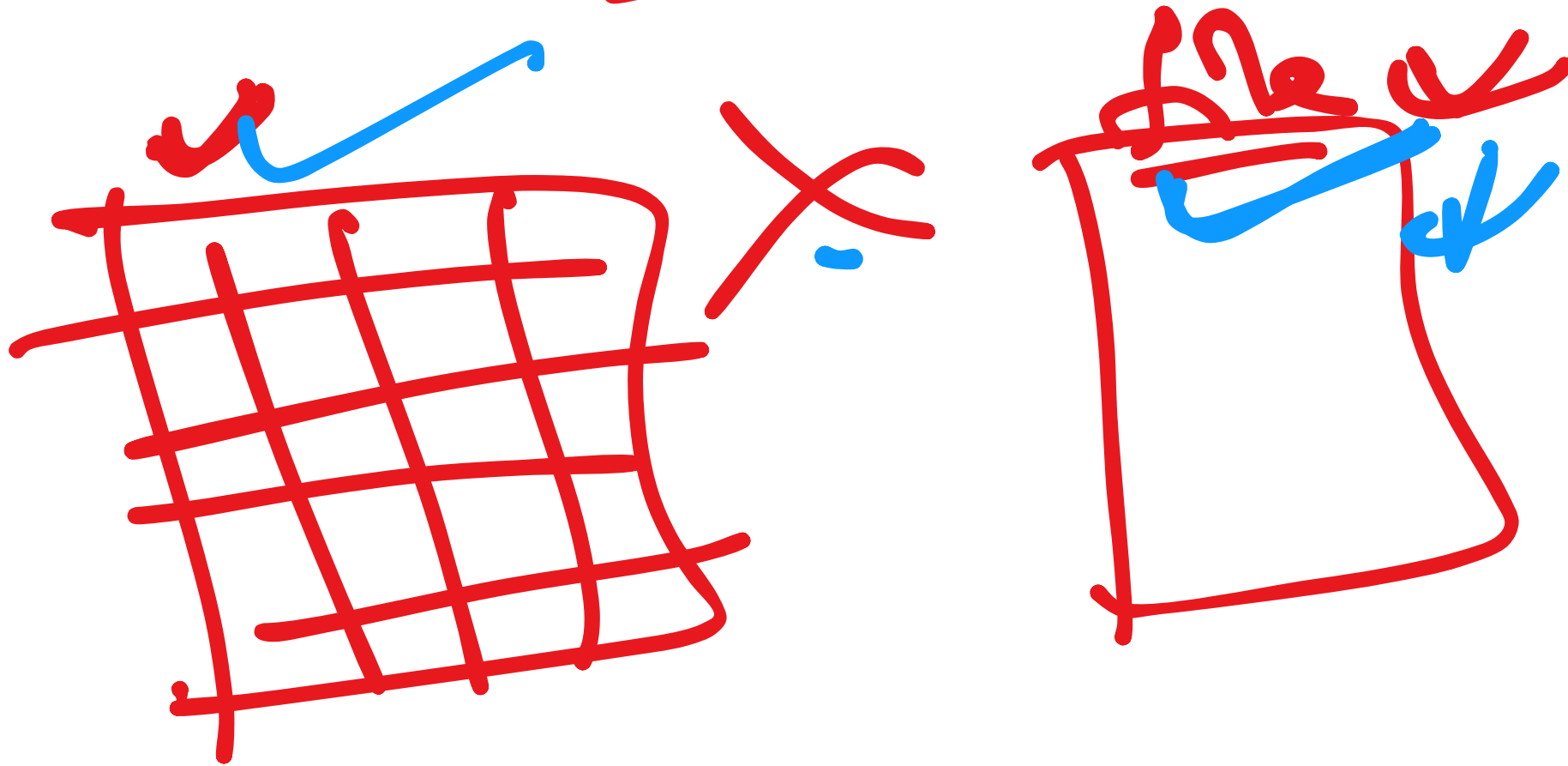
TABLE NAME	COLUMN NAME	DATA TYPE	COMMENTS
Doctor	ID	Number	Stores doctor ID
Doctor	Name	Character	Stores doctor name
Doctor	Area	Character	Stores doctor area
Doctor	Department	Character	Stores doctor's department

FLAT FILE

Flat files are simple files where data is not divided or structured properly. ✓

✓ They are easy to read but may be difficult to understand.

Examples: TXT, DAT, CSV, AVRO, Parquet, etc.

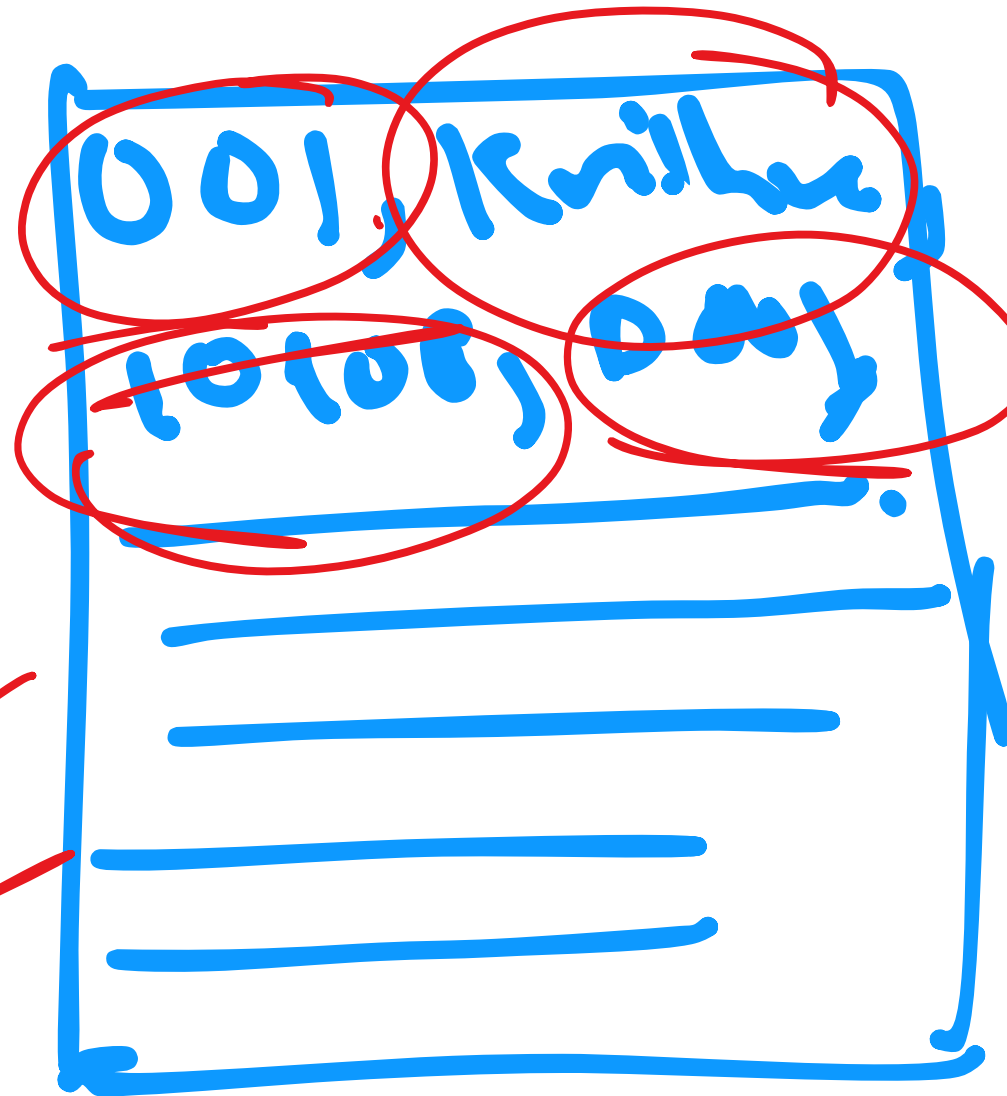


DELIMITERS

Delimiters are symbols used to separate data values.

Examples:

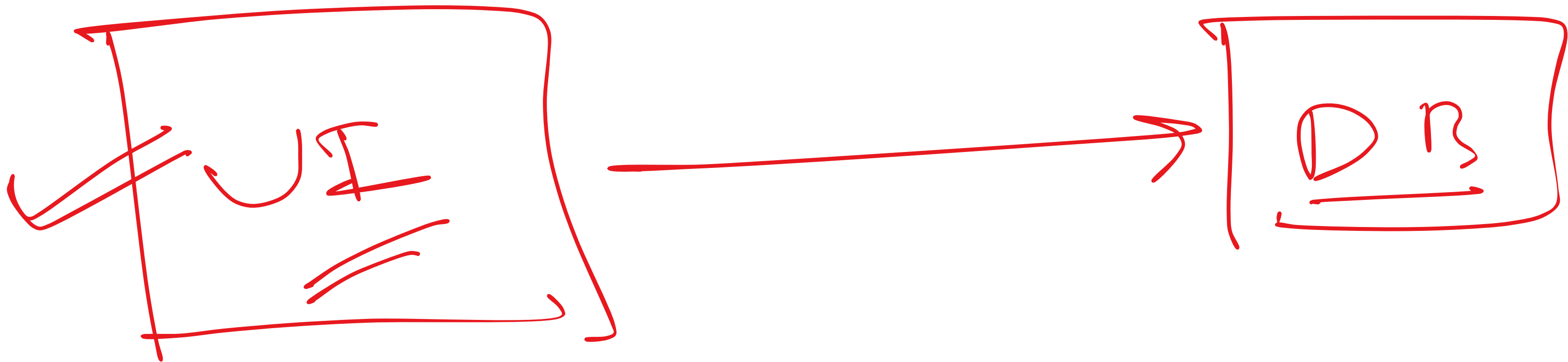
- **Comma (,)**
- **Dot (.)**
- **Semicolon (;)**
- **Tab**
- **Single pipe (|)**
- **Double pipe (||)**
- **Space**



FRONT END ✓

Front end is the part of software that users can see and interact with.

Example languages: C, C++, Java, .NET, VB, Python



BACK END

The back end is the server-side language used to store and process data.

Users cannot see this part — only back-end developers or data engineers work with it.

Example languages/tools: Oracle, MS SQL Server, DB2, MySQL

Note: In Data Warehouse projects, we mostly use back-end languages and BI tools.

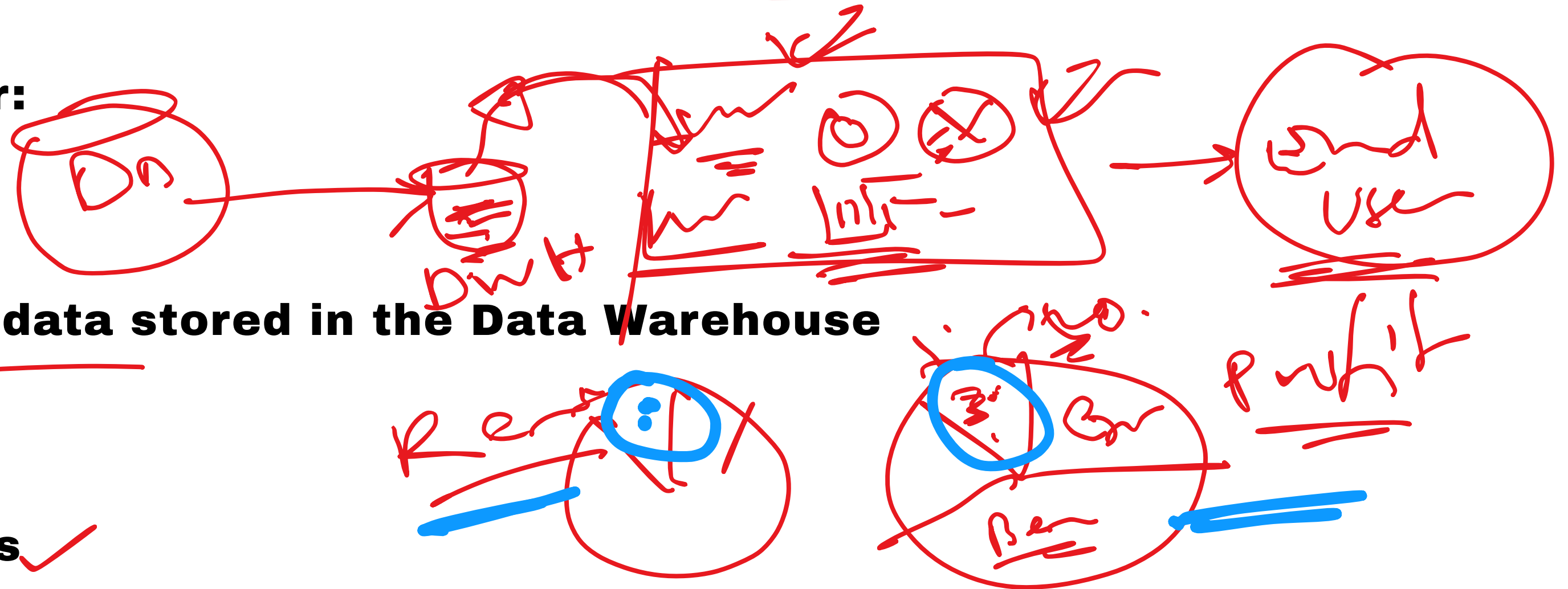


BI TOOLS (BUSINESS INTELLIGENCE TOOLS)

BI tools help collect raw or business data and convert it into meaningful information.

They are used for:

- Analysis ✓
- Reporting ✓
- Verification of data stored in the Data Warehouse

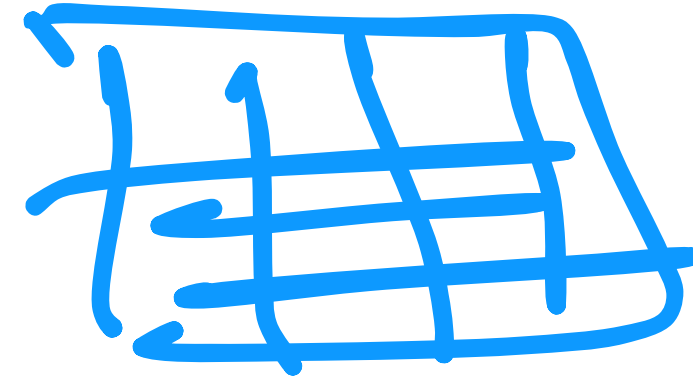


Common BI tools:

- Crystal Reports ✓
- OBIEE (Oracle Business Intelligence Enterprise Edition)
- MSTR (MicroStrategy)
- SSRS (SQL Server Reporting Services) ✓
- Tableau ✓
- Power BI ✓

DATABASE BASICS

Data in a database is stored in tables.



Different organizations may use different database languages such as:

- **Oracle**
- **MS SQL Server**
- **MySQL**
- **DB2**



A ~~database~~ is an organized way of storing and maintaining data on a computer.

DBMS (DATABASE MANAGEMENT SYSTEM)

DBMS is software that helps:

- Add data ✓
- Modify data ✓
- Delete data ✓
- Query the database ✓
- Generate reports ✓



Types of DBMS products:

- ✓ • Relational DBMS (multiple parents allowed)
- ✓ • Network DBMS (multiple parents)
- ✓ • Hierarchical DBMS (tree-like structure)

RDBMS (RELATIONAL DATABASE MANAGEMENT SYSTEM)

RDBMS stores data in the form of tables, fields, and records, and allows relationships between different tables.

This makes it easier to understand data and connect related information.

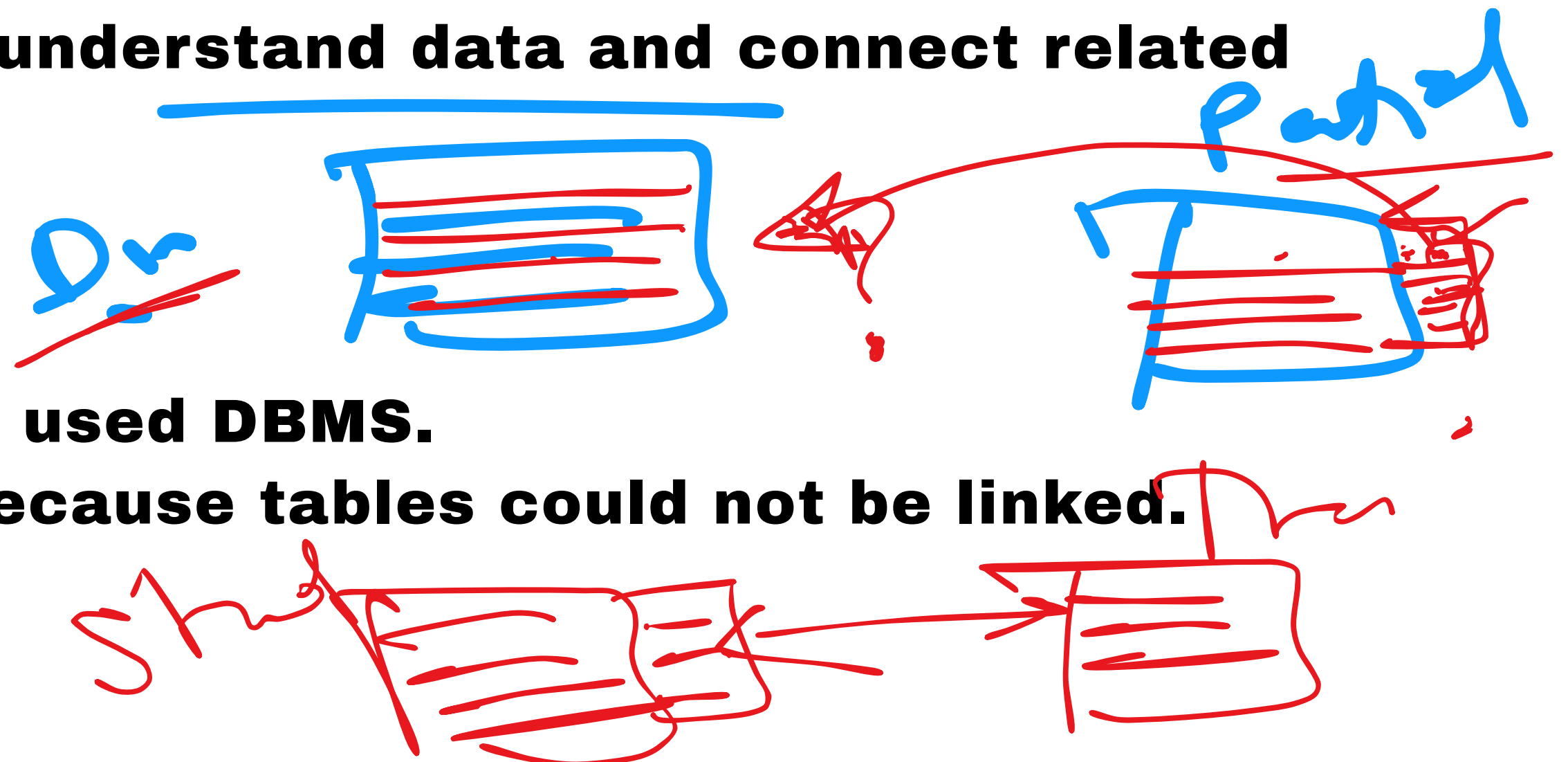
✓ Why RDBMS?

In the early 1970s, IBM used DBMS.

They faced problems because tables could not be linked.

For example:

- Doctor table
- Patient table



They could not find which patient was treated by which doctor. To solve this, IBM introduced RDBMS, which allows relationships between tables.

NEED OF A LANGUAGE

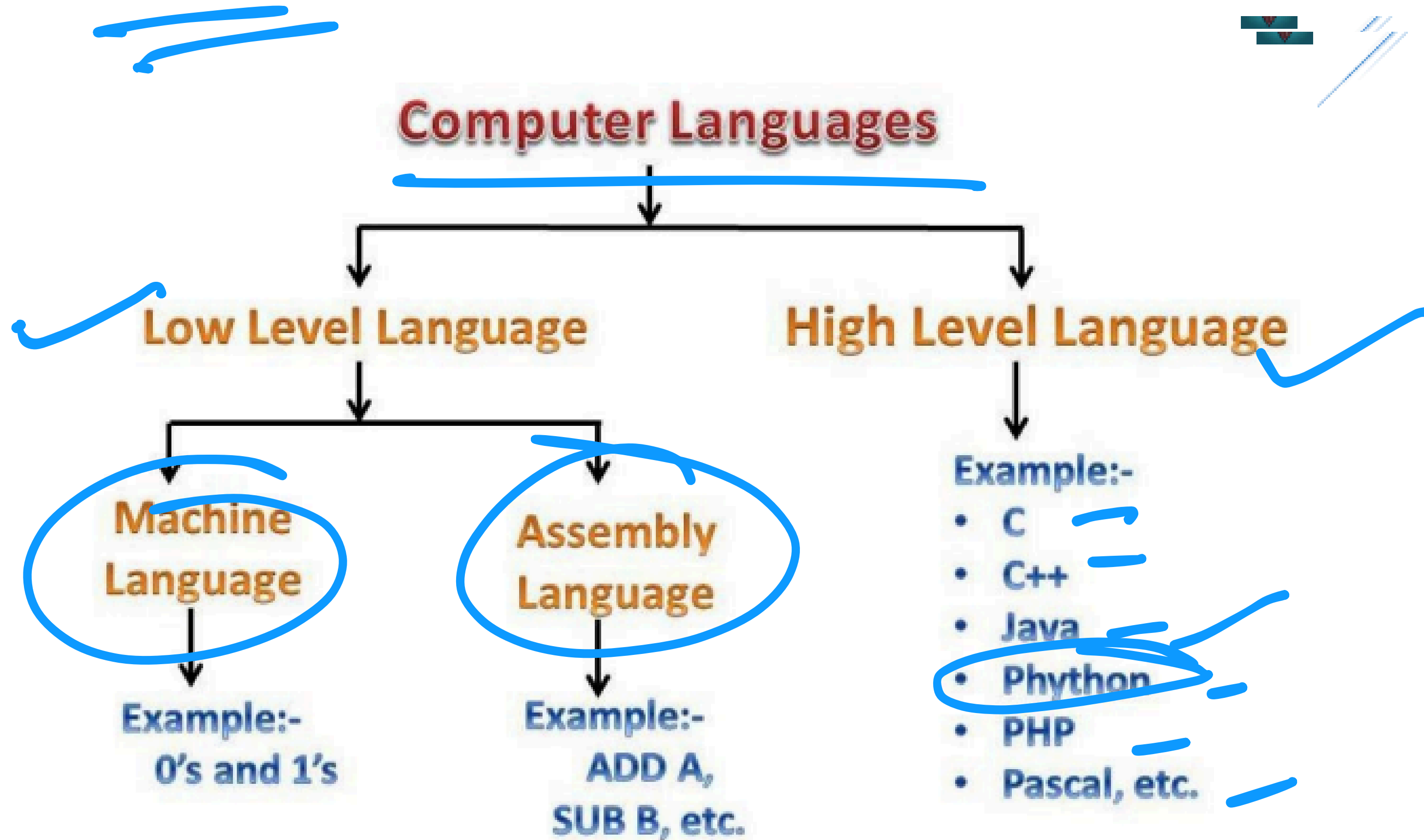
To communicate with databases, we need a common language.

Example analogy:

- User = Tap / Pump
- Language = Bucket (medium to transfer data)
- Database = Water tank
- Functions = Insert, Select, Delete commands



NEED OF A LANGUAGE



NEED OF A LANGUAGE

Only MS SQL

Note:

- Without a language, we cannot communicate with the database.
- All databases follow similar principles, but each company adds different features.

MySQL ✓

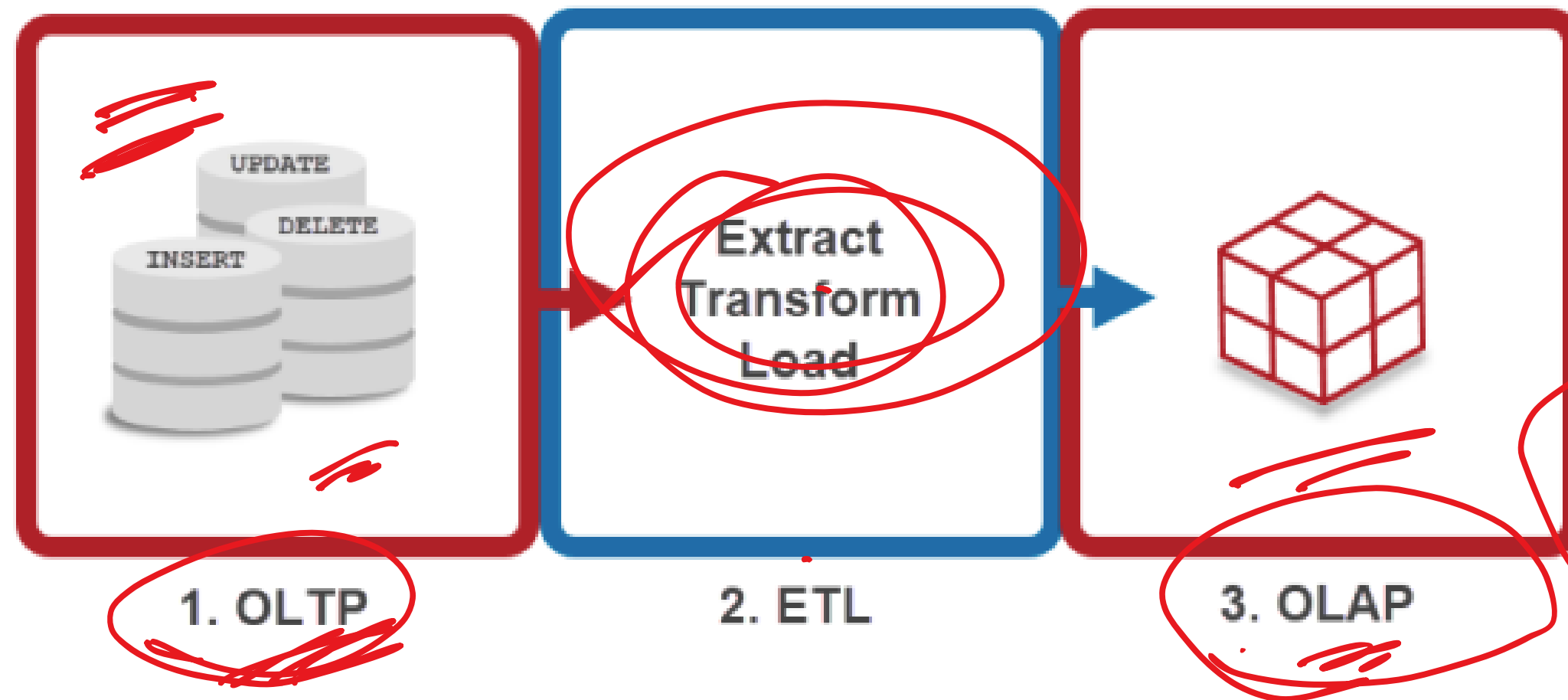
Oracle ✓

Q: Why do we need to upgrade databases?

A: For better performance and updated features.

WHY DO WE NEED A DATA WAREHOUSE?

To store data for long-term business analysis.
It keeps historical data that supports future decisions and reporting.



OLTP VS OLAP

OLTP → ONLINE TRANSACTION PROCESSING SYSTEM (SOURCE SYSTEM)

OLAP → ONLINE ANALYTICAL PROCESSING SYSTEM (TARGET SYSTEM / DWH)

ETL STANDS FOR EXTRACT, TRANSFORM, LOAD.

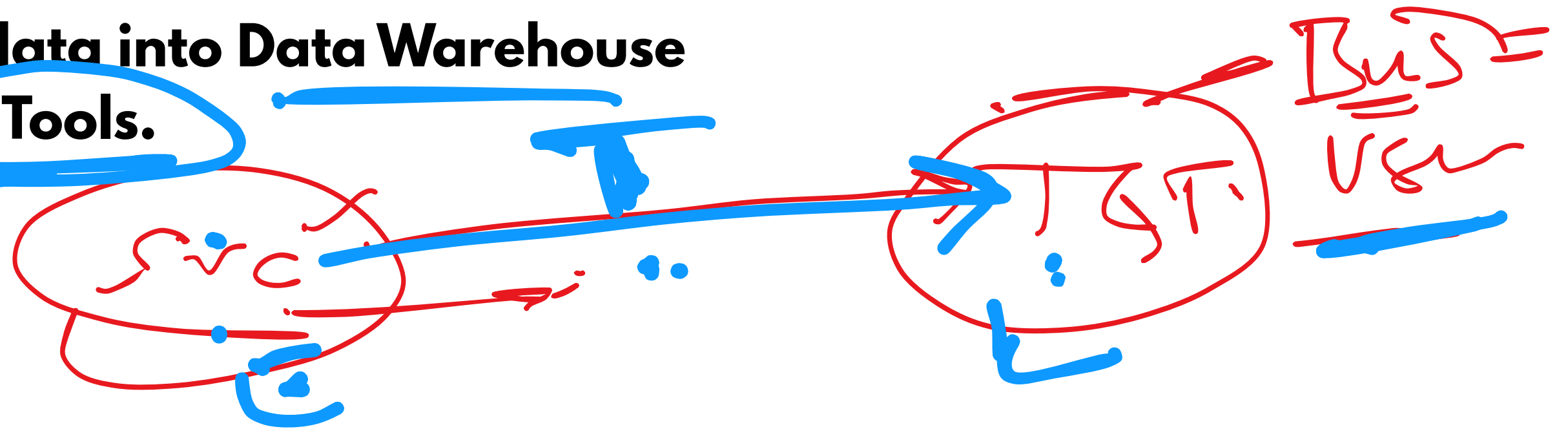
It is the process of moving data from OLTP to OLAP (Data Warehouse).

- Extract → Pull data from the source
- Transform → Convert data into standard format as per BRD (Business Requirements Document)
- Load → Store transformed data into Data Warehouse

This is implemented using ETL Tools.

Popular ETL Tools

- Informatica PowerCenter ✓
- Ab Initio ✓
- IBM DataStage ✓
- SSIS (SQL Server Integration Services) ✓
- Talend ✓
- ODI (Oracle Data Integrator) ✓

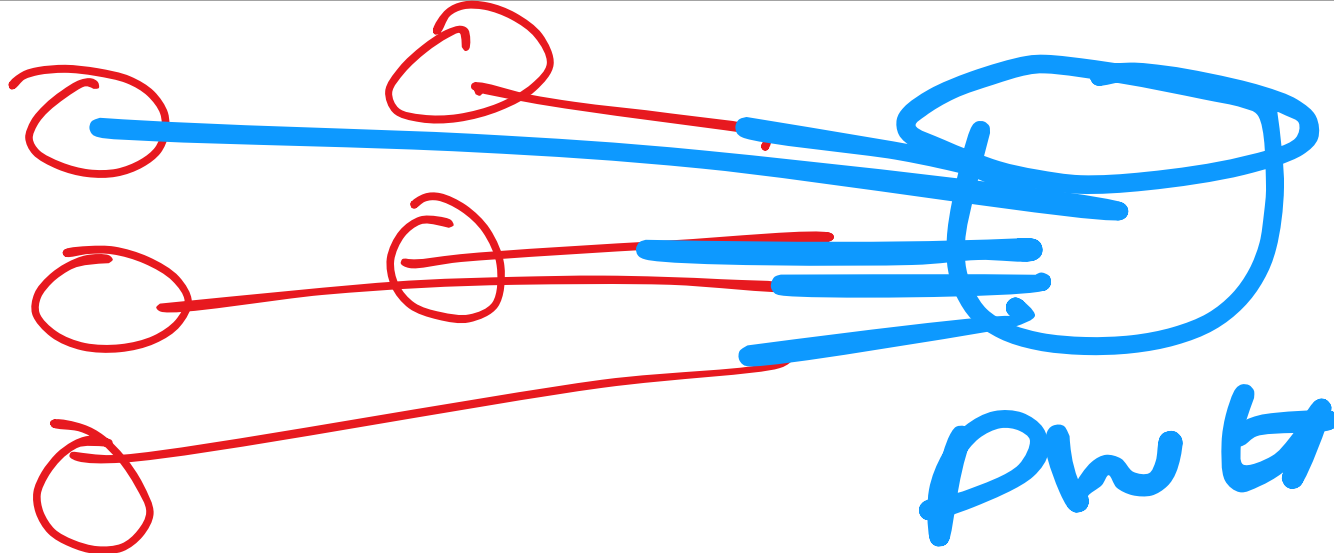


CHARACTERISTICS OF DWH

HITS

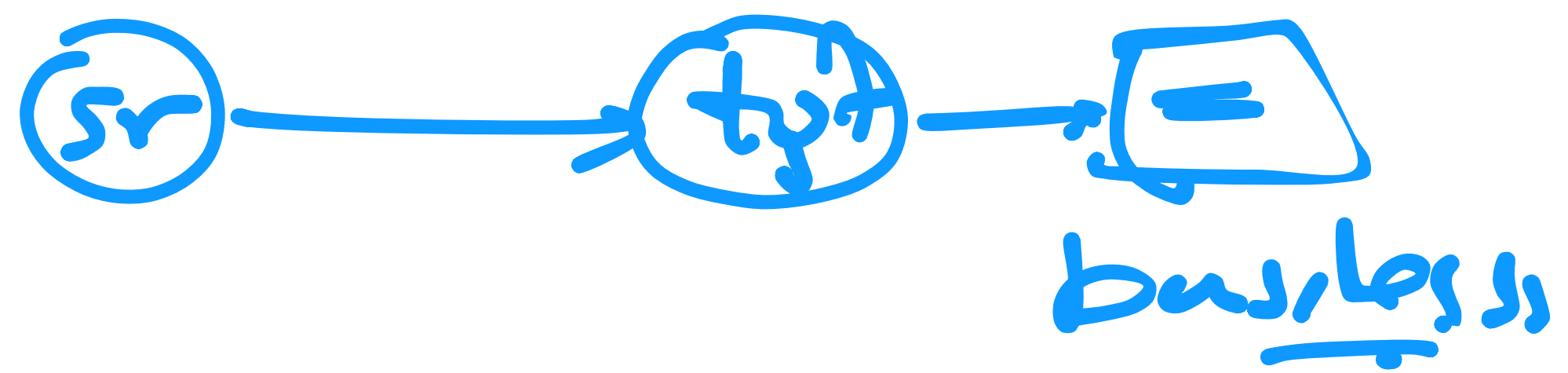
Feature	Meaning
✓ Subject Oriented	Organized by business <u>subjects</u> (<u>Sales</u> , <u>HR</u> , <u>Finance</u> etc.)
✓ Non-Volatile	<u>Once loaded</u> , data is <u>not changed</u>
✓ Integrated	Combines data from many <u>sources</u>
✓ Time Variant	<u>Stores historical data</u> (yearly, monthly, weekly etc.)

OLTP

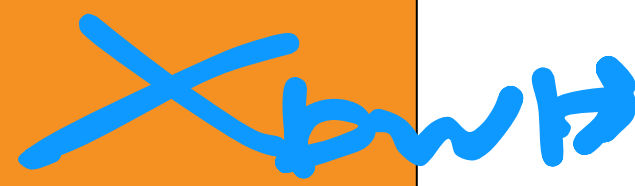


DIFFERENCE BETWEEN DATABASE (OLTP) & DATA WAREHOUSE(OLAP)

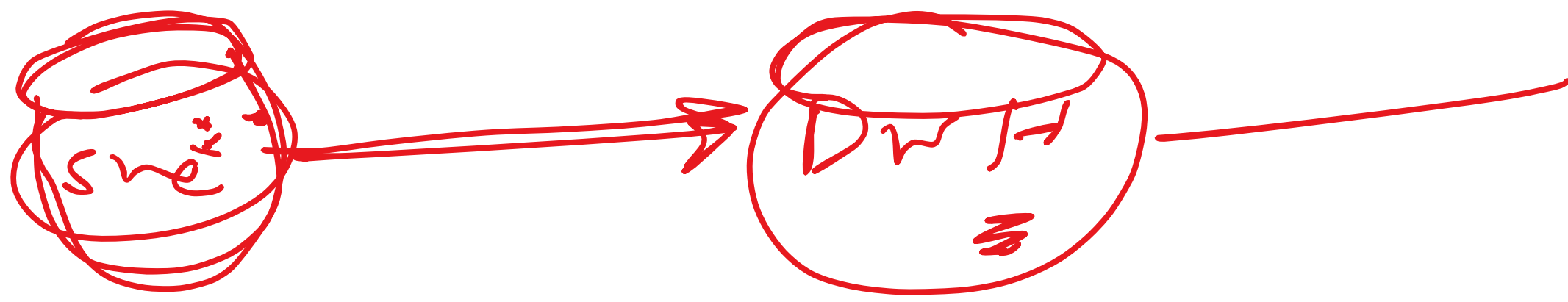
Data Warehouse (OLAP)	Database (OLTP)
Used for reporting	Used for transaction processing
Non-volatile (permanent)	Volatile (dynamic, updates happen)
Stores historical data	Stores current data
Stores additional timestamps with records	Data changes anytime
Subject-oriented, time-variant, integrated	Opposite characteristics of OLAP



DIFFERENCE BETWEEN DWH & WEB APPLICATIONS

DWH Project 	Web / <u>DB</u> Application Project 
Starts with <u>data</u> 	<u>Starts with requirement & ends with data</u> 
 Specially built <u>database</u> for <u>reporting</u> 	Not always a warehouse. 
OLAP 	OLTP 

Note: Every Data Warehouse is a database, but not every database is a Data Warehouse.



NORMALIZATION VS DE-NORMALIZATION

Normalization	De-Normalization
Used in OLTP ✓	Used in OLAP ✓
Reduces <u>redundancy</u> ✓	Improves performance ✓
Saves storage space ✓	<u>Faster data retrieval</u> ✓
Based on <u>ER Model</u> ✓	Based on <u>Dimensional Model</u>

Redundancy
Splitting

BEFORE: EMPLOYEE DATA ✓
| EMPNO | ENAME | SAL | DNO | DNAME | LOC |

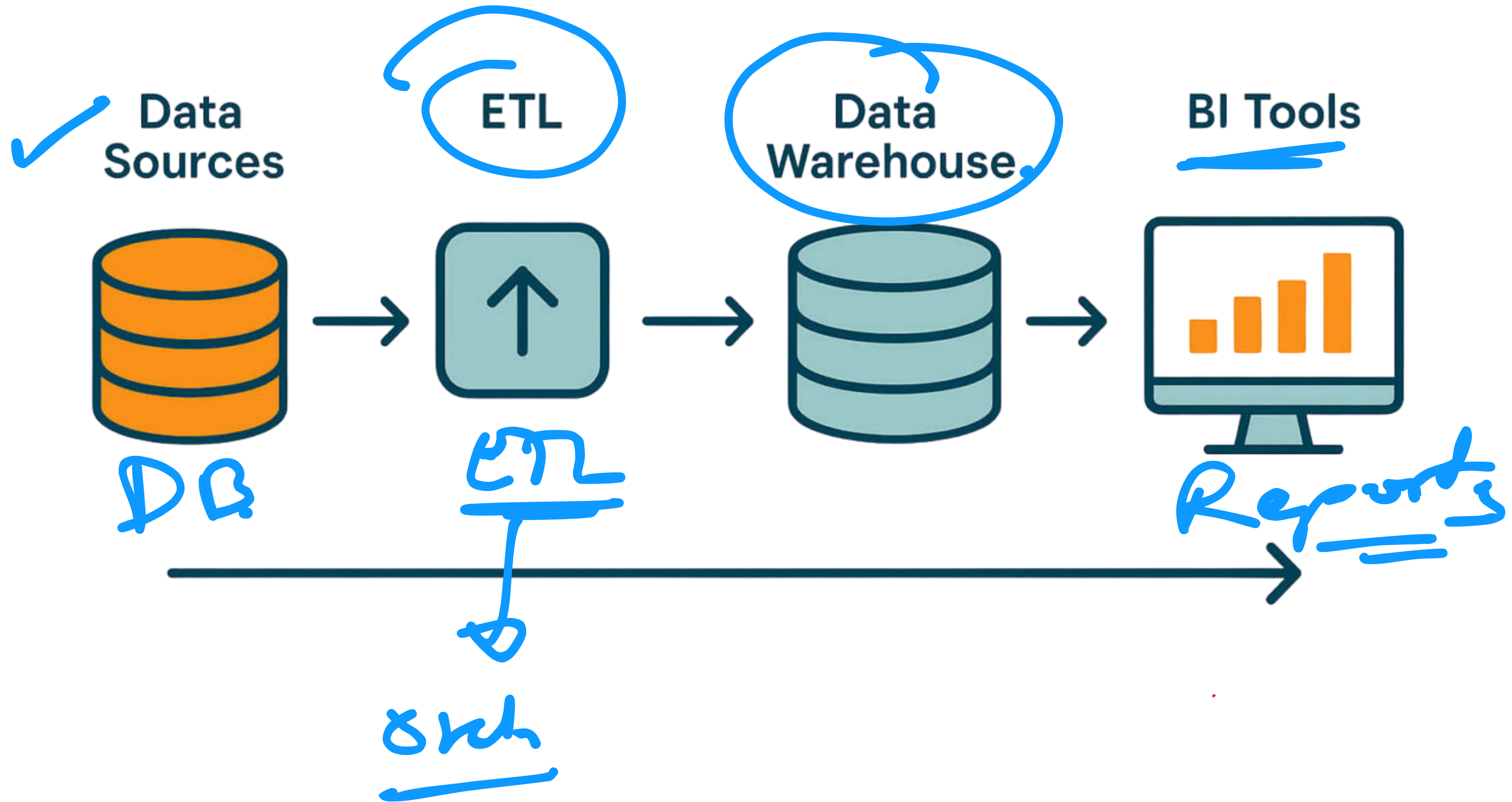
Query fast

OLAP

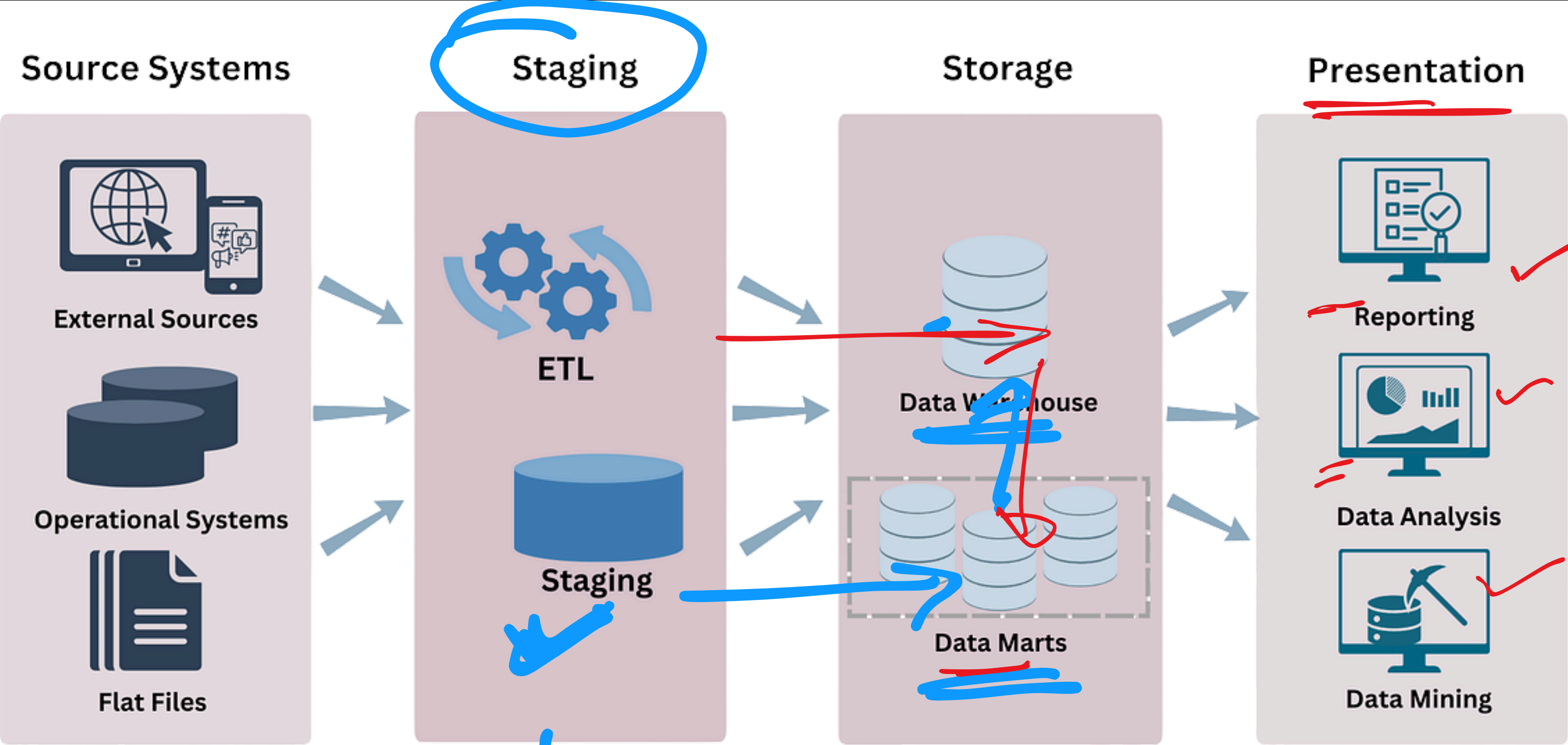
AFTER SPLITTING:
EMPLOYEE - | EMPNO | ENAME | SAL | DNO |
DEPT - | DNO | DNAME | LOC |

① } OLTP
② }

ARCHITECTURE OF DATA WAREHOUSE



ARCHITECTURE OF DATA WAREHOUSE



ARCHITECTURE OF DATA WAREHOUSE

1 ☒ SOURCE → STAGING (STG) → DWH → REPORTS

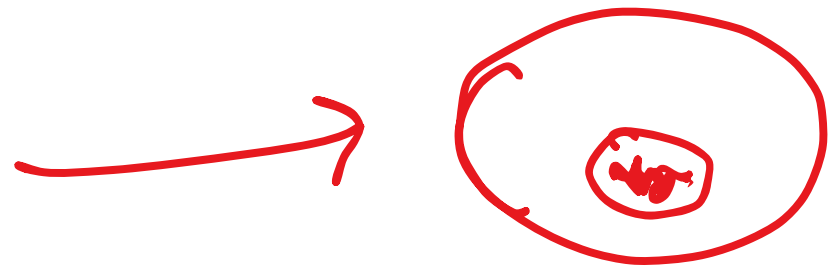
2 ☒ SOURCE → STG → DATA MART → REPORTS

3 ☒ SOURCE → STG → DWH → DATA MART → REPORTS

4 ☒ SOURCE → STG → DATA MART → DWH → REPORTS



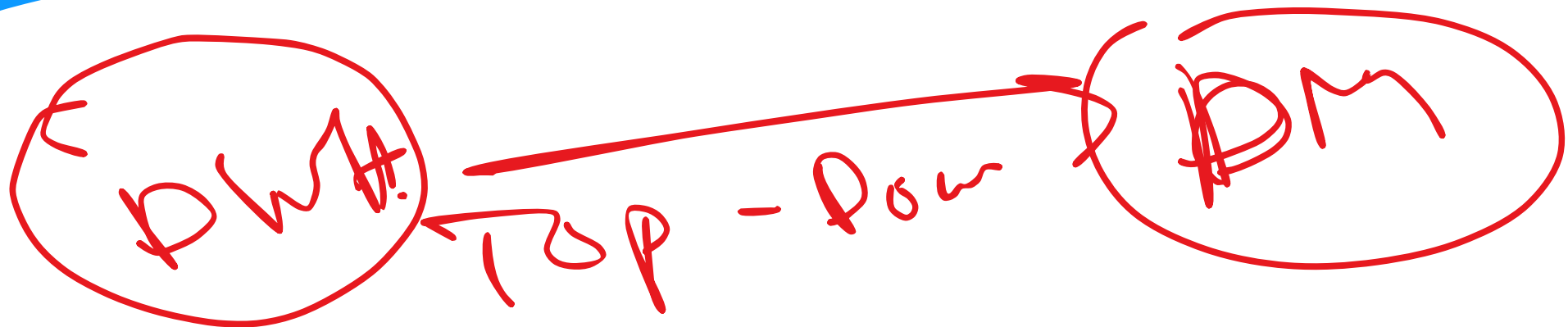
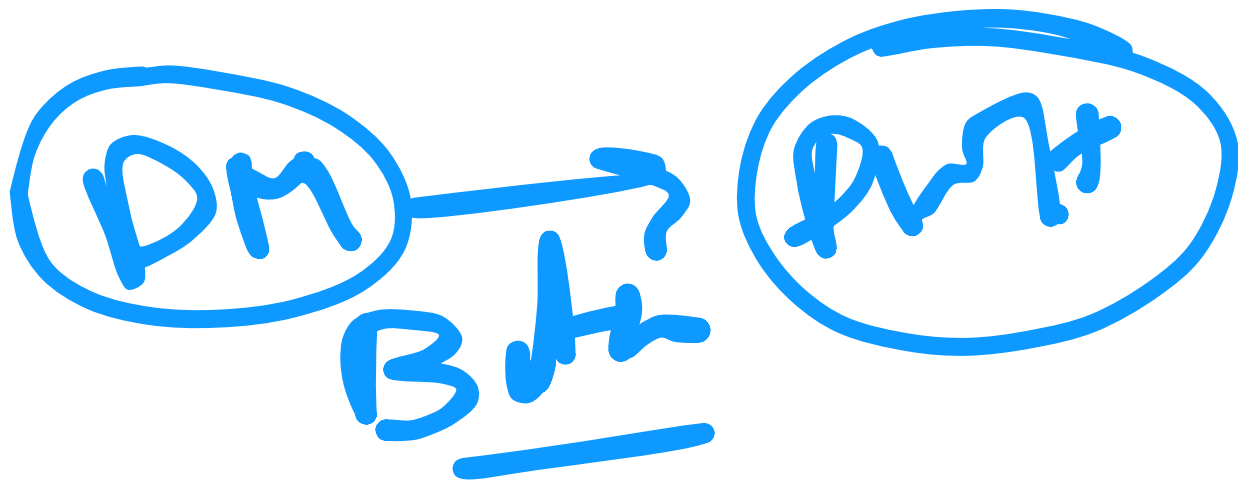
DATA MART



Top-Down Approach	Bottom-Up Approach
Data Warehouse built first	Data Marts built first
DMs depend on DWH	Later integrated to form DWH

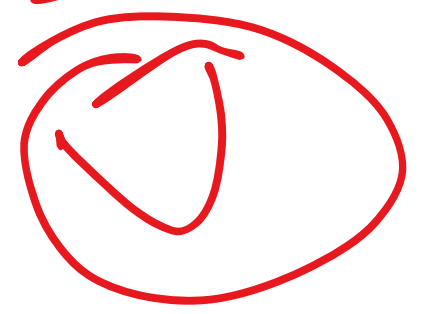
Types of Data Marts

- Dependent (uses data from DWH)
- Independent (directly from sources)



DIFFERENCE BETWEEN DATA WAREHOUSE & DATA MART

Data Warehouse	Data Mart
Collects data from <u>many sources</u>	<u>Few sources</u> 
Enterprise-wide 	Subject-specific 
<u>Large volume</u> → <u>Slow to query</u>	<u>Small volume</u> → <u>Fast reporting</u>

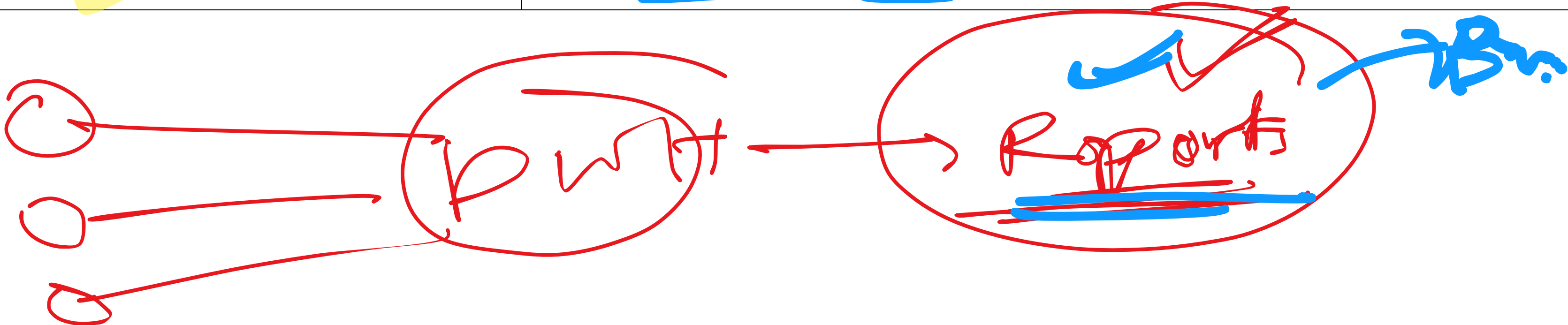


DATA MODELS

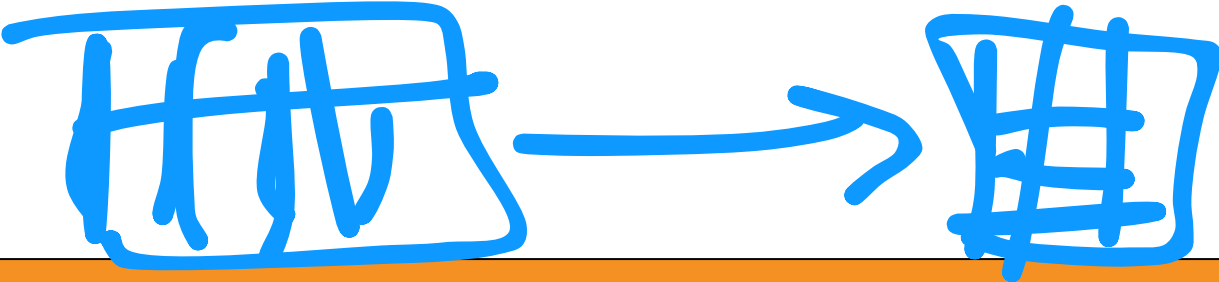
~~CLP~~ D.D.D.D.D.D.D.D.

Data Model	Description
Conceptual Model ✓	High-level <u>entities</u> & <u>relationships</u> (no attributes/keys)
Logical Model ✓	Defines all <u>attributes</u> & <u>keys</u>
Physical Model ✓	<u>Actual</u> <u>tables</u> , <u>columns</u> , <u>data</u> <u>types</u>

↓



DIMENSIONAL DATA MODEL

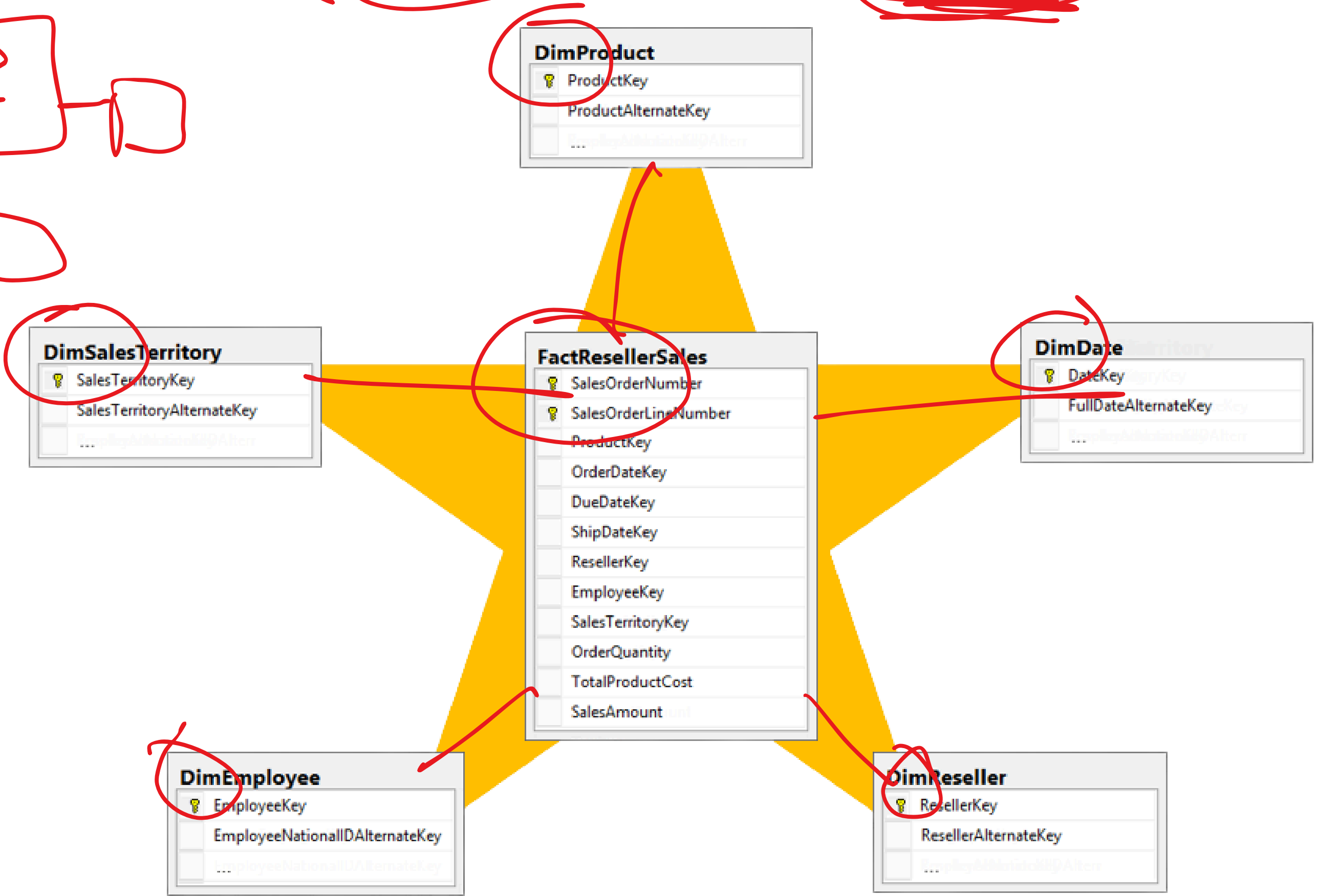
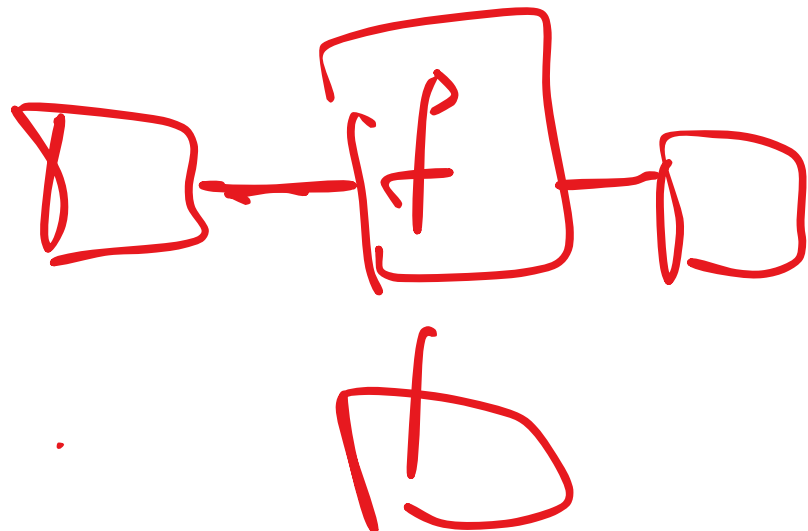


Fact Table ✓	Dimension Table ✓
Contains numerical, summarized data	Contains descriptive, detailed data
Child table (foreign key)	Parent table (primary key)

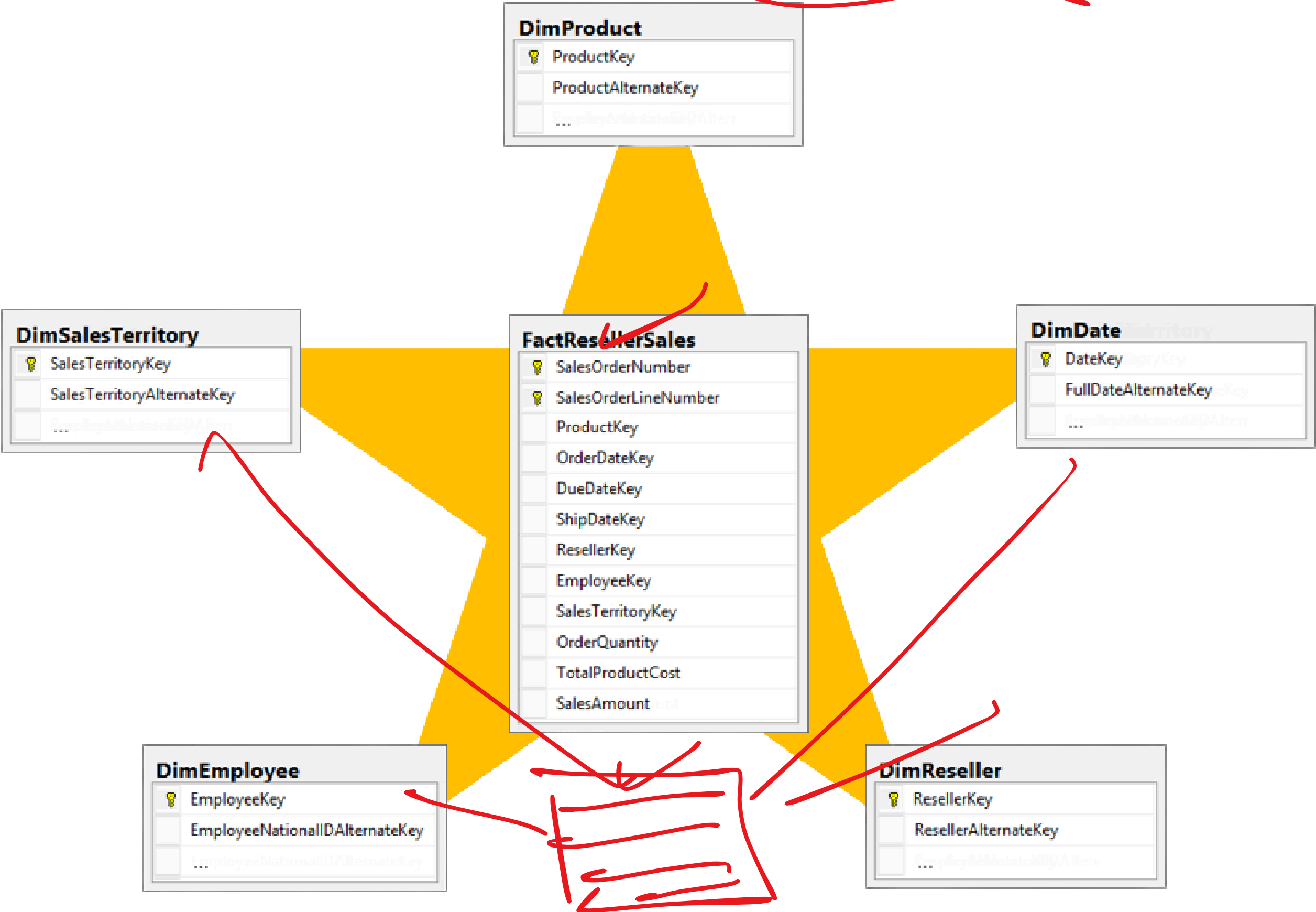
Factless Fact Table:

A fact table without any fact measure.

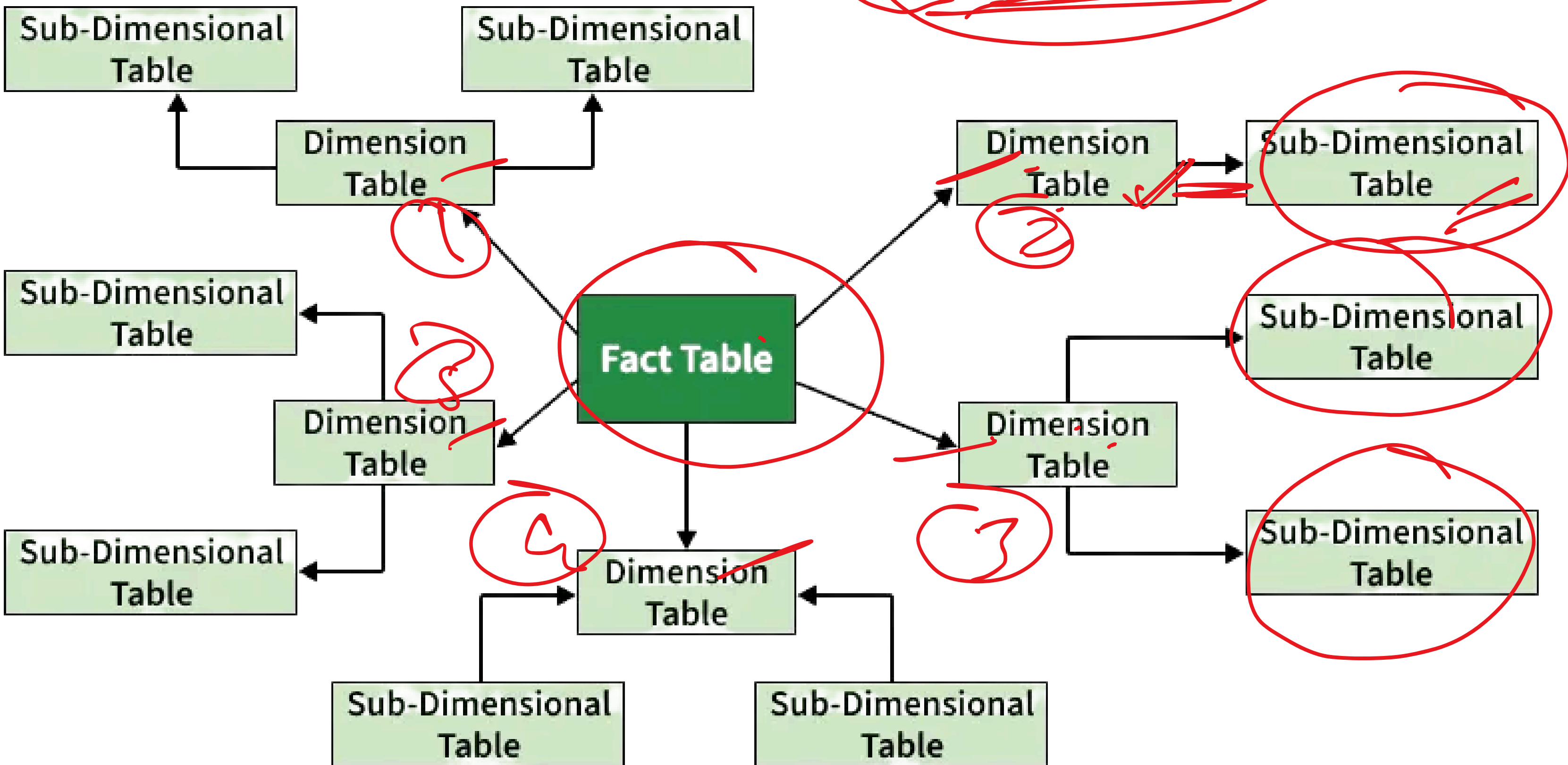
DIMENSIONAL DATA MODEL - STAR SCHEMA



DIMENSIONAL DATA MODEL - COMPLEX STAR SCHEMA

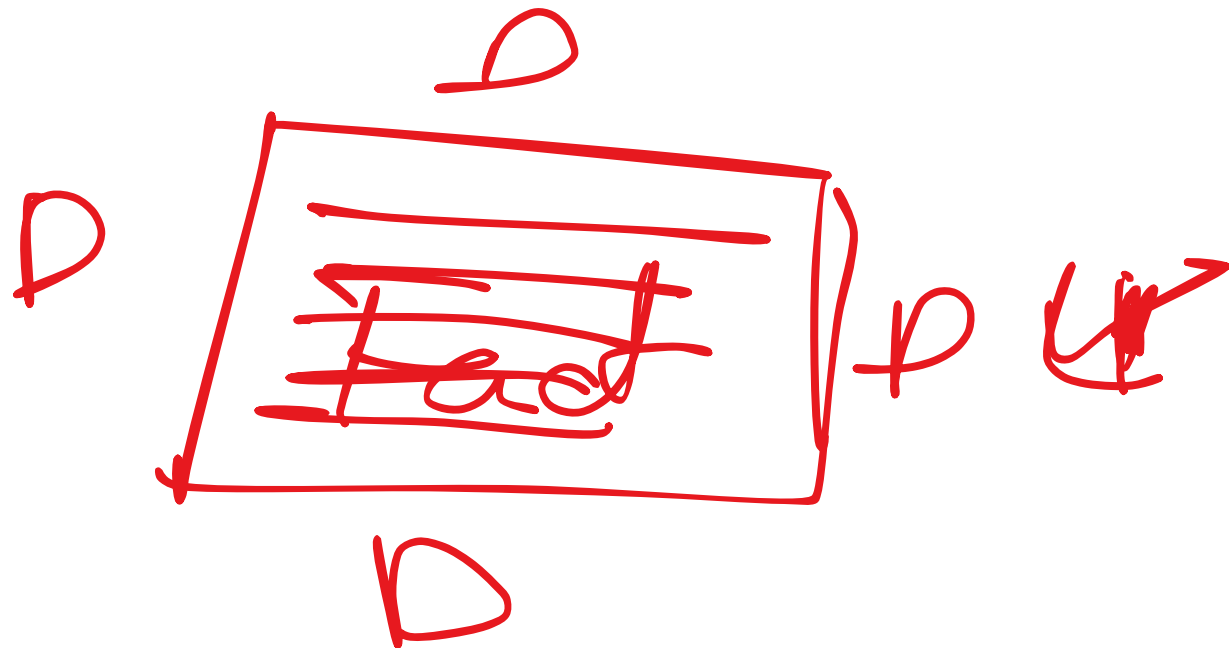


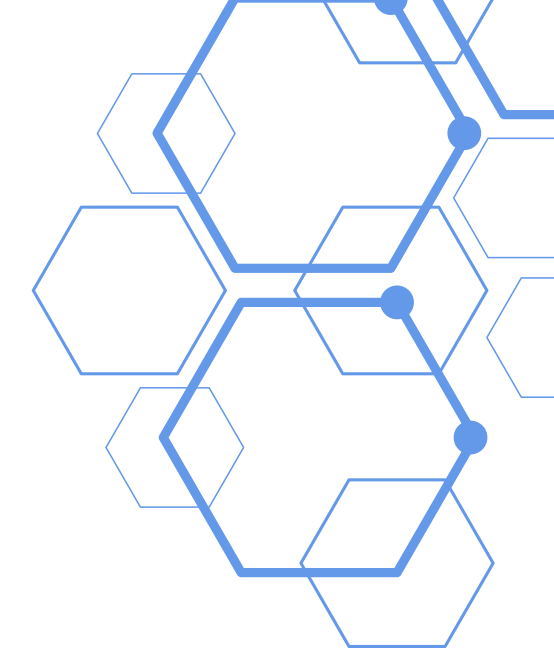
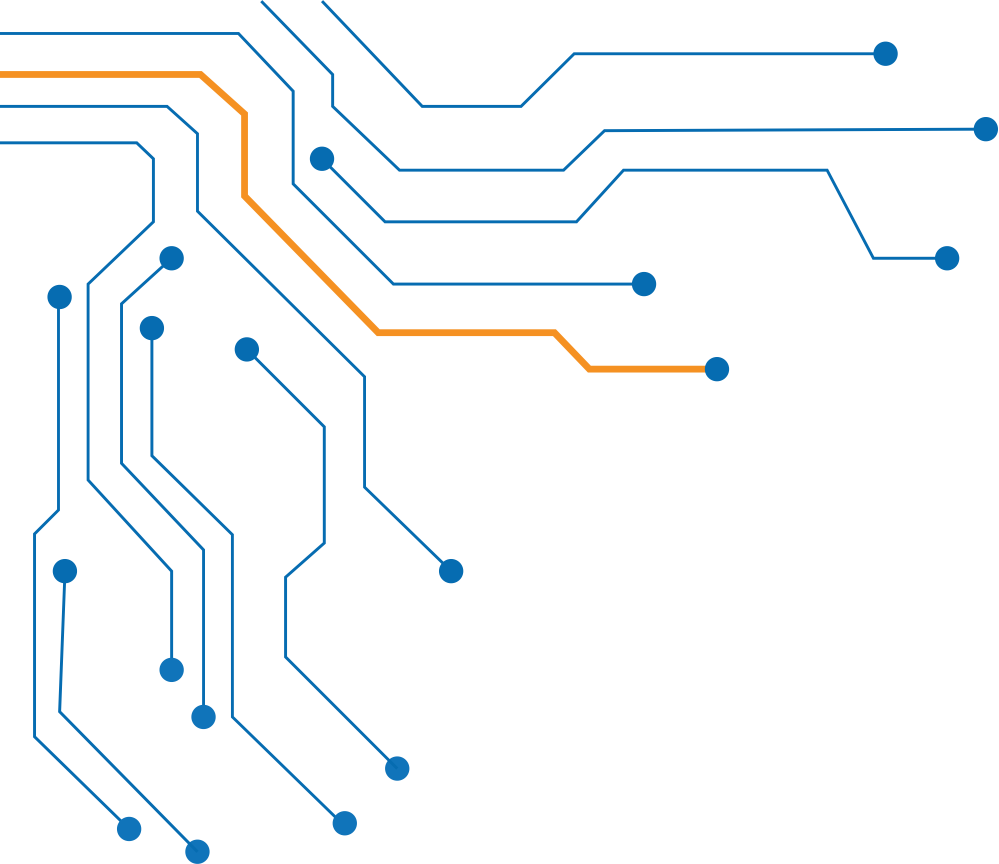
DIMENSIONAL DATA MODEL - SNOWFLAKE SCHEMA



IMPORTANT NOTES

- Most real-time projects use Star Schema
- Dimension = Walls of a room
- Fact = Roof of a room
- Dimensions are independent, Facts are dependent





DATAWARE HOUSE CONCEPTS

GENERAL TERMINOLOGIES IN DATABASE AND DATA WAREHOUSE

